

Qatar-1b

R-1419

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210422 · created 2026-08-01T06:05:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459326.9158 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.011
Transit depth (Rp/Rs)^2	2.56 +/- 0.37 [%]
Orbital Inclination (inc)	84.99 +/- 0.93
Airmass coefficient 1 (a1)	1.306 +/- 0.018
Airmass coefficient 2 (a2)	-0.169 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	None
Optimal Aperture	3.44
Optimal Annulus	10.37
Transit Duration (day)	0.0742 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.011 vs 0.1463 (deviation 1.25σ)
Duration fit vs published	0.0742 d vs 0.0692 d (deviation 1.23σ)
Mid-time O–C	-0.8 min
Depth SNR	14.5
Residual scatter	1.38 %

Light curve

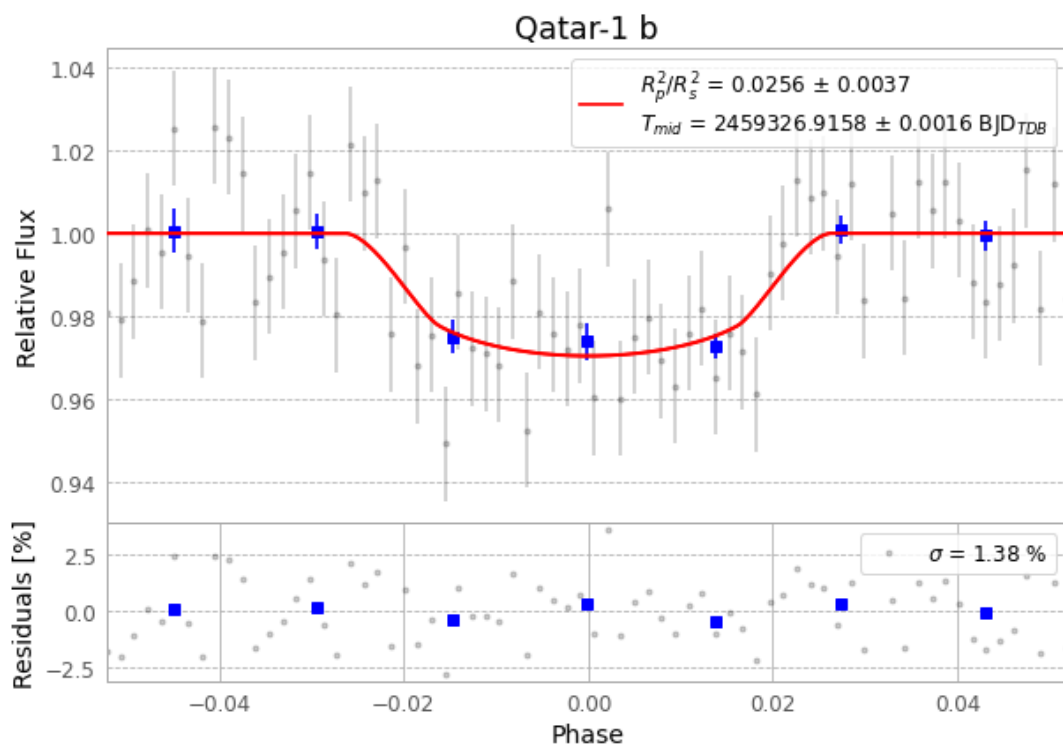


Figure 1 — FinalLightCurve_Qatar-1 b_2021-04-22.png (SHA-256 47819674c3705005...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 efb41062e8b1192d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), Qatar-1210422081212.FITS → Qatar-1210422114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210422.log (e81e18f1befb...), FinalLightCurve_Qatar-1 b_2021-04-22.png (47819674c370...), FinalLightCurve_Qatar-1 b_2021-04-22.csv (5f05a95b9539...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:05Z.